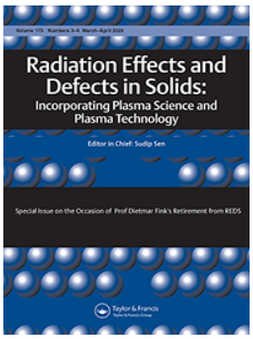

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Materials parameters and ion-induced track formation

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ABSTRACT
The role of various materials parameters in track formation is discussed. Experimental information is utilized showing a direct quantitative relationship in different solids between the melting points T_m and the ion-induced track radii R_e without involving other materials parameters. It is shown for $\Theta = T_m - T_{ir}$ (T_{ir} – irradiation temperature) that $\Theta \sim \exp\{-R_e{}^2/w^2\}$ for $S_e/N =$ constant, where S_e and N are the electronic stopping power and the number density of atoms and $w = 4.5$ nm. The validity of this universal-type relation is demonstrated for 14 different insulators including $LiNbO_3$ and $BaFe_{12}O_{19}$. It is shown that the thermal diffusivity D, the heat of fusion L the band gap energy E_g and the absorption radius α_e of the electron distribution must not affect the track sizes as this would not be coherent with this identical behavior. Original reports on $LiNbO_3$ and $BaFe_{12}O_{19}$ with opposite conclusions are critically analyzed. It is shown that an arbitrary value of the ion energy was used in the analysis that modified substantially the results leading to an undue justification of the contribution of L and E_g in apparent agreement with the experimental data.

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Heavy ions; tracks; velocity effect; heat of fusion; band gap energy

1. Introduction

In spite of the considerable progress in the studies of the electronic processes in ion-solid interaction in the last three decades, even some simple questions cannot be answered unambiguously at present. Such problem is the fraction of the electronic stopping power S_e that is transferred to the thermal energy ϵ of the ion-induced spike. Another fundamental problem is which are those individual materials parameters (MPs) that affect the track formation? Various models offer different answers to these questions.

It is evident that this is a very important problem both from theoretical and practical aspects as well. It is sufficient if one considers only the various solids that are applied in the nuclear industry for any purpose. The sensitivity of the radiation response of these solids to various MPs may be critical for applications. On the other hand, the knowledge of the right MPs controlling this response is a highly important element for elaborating the valid model.

There are essentially two main positions: (i) track sizes depend on many MPs, (ii) track sizes do not depend on MPs but one. When experiments are analyzed a number of MPs

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and fitting parameters are applied according to (i) and only a single MP without any fitting parameter is used according to (ii). However, satisfactory description is provided by (i) only when an incomplete analysis is also accepted that does not account on the scaling features of the track data though it has been reported already by about 25 years (*1*). Thus the quantitative relationship between the results in different solids is ignored.

In this paper, we renew the efforts for the clarification of this problem with MPs by applying a general approach completed with checking the information on some 'classical' experiments in this field which are referred frequently in this respect for supporting (i). In this paper, only the problems related to the thermal diffusivity D, the heat of fusion L, the band gap energy E_g, and the absorption radius α_e are discussed. A peculiarity of the analysis is that opposite conclusions are drawn in the literature from the same experimental data regarding the role of L and E_g. In this paper, we discuss both reasonings. After reviewing the general features of track formation, they are applied to the problem of the contribution of various MPs to track formation and earlier studies are critically analyzed.

2. Discussion

2.1. Scaling features of track evolution

When an insulator is irradiated by swift heavy ions often amorphous cylindrical tracks are formed along the projectile trajectory. In a simple case, when the specific ion energy $E =$ constant, the tracks are characterized by the R_e radius that is a monotonous function of S_e. Otherwise, R_e is sensitive to the variation of E: the track size increases considerably with the reduction of E at $S_e =$ constant. This is the so-called damage cross-section velocity effect (VE) (*2*). Throughout this paper $S_e >> S_n$ is assumed where S_n is the nuclear stopping power characterizing the elastic interaction.

It is a specific property of track formation that when ${R_e}^2$–S_e track evolution curves are compared in various insulators a scaling feature is observed. In a simple case, the scaling factor is given by the number density of atoms N and the melting temperature T_m (*3*). This is rather unusual in experimental physics. In case of a similar behavior this might mean that the variation of the electrical or magnetic or mechanical or optical properties with temperature or pressure in a solid it can be obtained within experimental error by measurements on another solid. Such a possibility exists for track formation and this makes much easier answering to the problems formulated in the introduction.

In Figure 1, typical ${R_e}^2$–S_e track evolution curves are shown. Track formation starts for $S_e > S_{et}$ where S_{et} is the threshold electronic stopping power that can be estimated from the intersection of the lines with the X-axis. The semilogarithmic lines are parallel with the slope $d{R_e}^2/d\ln S_e \approx 20.25\,\mathrm{nm}^2$. Tracks were induced in $Y_3Fe_5O_{12}$, by low velocity (*2*, *4*), and high velocity ions (*2*), as well. They are clearly separated in this plot showing that LO ($E < 2$ MeV/nucleon) ions induce considerably larger tracks than HI ($E > 8$ MeV/nucleon) ions at the same values of S_e. This plot is a nice demonstration of the VE. The next figure exhibits the closest quantitative relationship between the track data.

In Figure 2, track radii and melting temperatures are shown for various track forming insulators for irradiations by low and high velocity ions when the atomic stopping power $< s_e > = S_e/N =$ constant. According to the figure, there is a simple quantitative relation

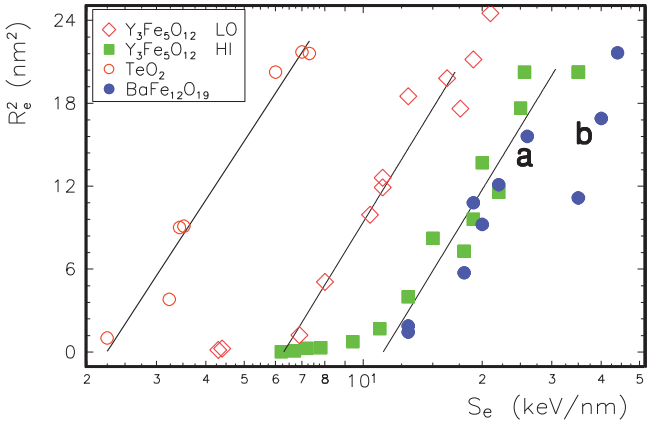

Figure 1. Variation of the track size (R_e – track radius) with the electronic stopping power S_e in $Y_3Fe_5O_{12}$; irradiation by low ($E < 2$ MeV/nucleon, LO) and high velocity ions ($E > 8$ MeV/nucleon, HI) – (*2*, *4*), TeO_2 (*5*) and $BaFe_{12}O_{19}$ (*6*, *7*).

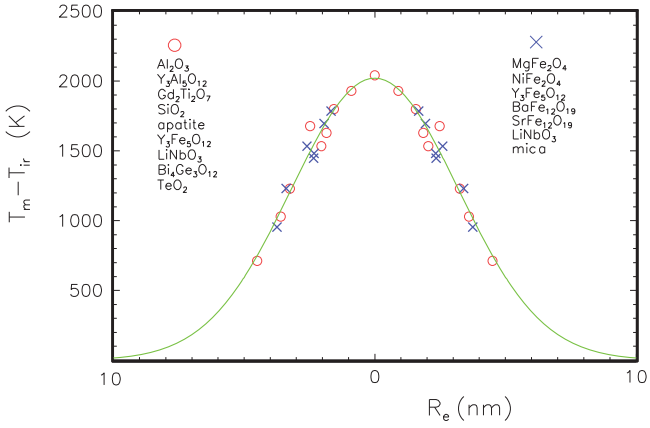

Figure 2. Variation of the track radii R_e with the melting temperature T_m in insulators irradiated by low (o) ($E < 2$ MeV/nucleon) and high velocity (x) ($E > 8$ MeV/nucleon) ions at $< s_e > /3k = 3.42 \times 10^5$ nm^2 K and 7.5×10^5 nm^2 K, respectively (T_{ir} – irradiation temperature, k – Boltzmann constant). In the figure, the solids are listed in the order of decreasing melting temperatures T_m. The enveloping curve $\Theta(r) = T_p \exp\{-r^2/w^2\}$ is a fit with $w = (4.5 \pm 0.3)$ nm and $T_p = 2020 \pm 60$ K (for track data see (*8*)).

between track radii induced in different solids by different ions with different E and S_e values in the LO or the HI ranges when $< s_e > /3k$ has fixed values where k is the Boltzmann constant.

The existence of this kind of relationship is itself extraordinary. It means that if we do not know any of the track data in Figure 2 and measure a single track, *e.g.* in mica induced by Xe ions with $E = 11.4$ MeV/nucleon then this would be sufficient information (completed with the appropriate T_m values) for the calculation of all track radii in the figure within experimental error. We emphasize that such relationship between solids is valid only for track sizes. Any other properties of the solids in Figure 2 are independent. It is evident that the

origin of this extraordinary effect must be related to the basic features of track formation in these solids. And there must be considerable discrepancies from those schemes which are incompatible with this feature, *e.g.* from those theories which are based on the contribution of many MPs.

Actually, in Figure 2, the R_e – $(T_m - T_{ir})$ relation (T_{ir} – irradiation temperature) is described by the same $\Theta(r)$ function in the LO and HI ranges (*8*)

$$\Theta(r) = T_p \exp\{-r^2/w^2\} \tag{1}$$

where $w = 4.5$ nm and the peak temperature is $T_p = 2020$K under the given experimental conditions. In the figure, the values of $< s_e >$ were chosen with the purpose for having equal T_p peak temperatures in the LO and HI ranges. It is remarkable that the shape and the width of $\Theta(r)$ are not modified with varying E, only T_p may vary. This is a highly significant result.

An extremely important information provided by Figure 2 is that the track sizes are controlled only by the melting temperature T_m. This confirms unambiguously the thermal mechanism of track formation. Though there are data for 14 different insulators in Figure 2 and values of their MPs certainly vary in broad ranges, nevertheless, the track radii follow closely the same curve described by Equation (1). This is an indisputable evidence demonstrating that other MPs apart the melting temperature do not affect the track radii within experimental error. This information is of fundamental importance for understanding the irradiation effects.

In a thermal spike, amorphous tracks are formed from a cylindrical melt and the track diameter $2R_e$ is equal to the maximum width of the ion-induced temperature distribution at the melting temperature. $T(r,t)$ is the ion-induced temperature increase and its value is $T_m - T_{ir}$ when the local temperature is T_m. The time is measured starting from the moment of attaining the maximum width at the melting point at $t = 0$ when the temperature increase is $T(r,0)$. By using $\Theta(r)$ this width – the track diameter – can be obtained at the appropriate melting temperatures for any solid in Figure 2 as $T_m - T_{ir} = \Theta(R_e) = T(R_e,0)$. It is remarkable that $\Theta(r)$ is a simple analytical function without individual MPs, including the parameters of the melting of various solids. These widths are all closely related to each other by means of $\Theta(r)$ as any of them can be estimated from any other one. Consequently, the $T(r,0)$ distributions must be also quantitatively related to $\Theta(r)$ and to each other in these solids. This excludes the possibility that any of the $T(r,0)$ distribution may depend on individual MPs, as well. They may depend on experimental parameters $< s_e >$ and T_{ir}. However, these parameters are identical in Figure 2. Finally, the track data are sorted into two groups as those with $E < 2$ MeV/nucleon and $E > 8$ MeV/nucleon in agreement with the results of previous analyses (*3*) and Figure 2 confirms such selection.

If the $T(r,0)$ distributions do not depend on individual MPs and there exists a mathematical relationship between them as discussed above, then they must be identical. Evidently, this conclusion is valid for all 14 insulators in Figure 2, as well. As the common $T(r,0)$ and $\Theta(r)$ are equal at all 14 melting temperatures, consequently, they can be considered identical. A more detailed discussion has been given in Ref.(*8*). Thus the $\Theta(r)$ distribution is the analytical description of the ion-induced temperatures in insulators. This explains that $\Theta(r)$ provides simultaneously the correct widths for all insulators in Figure 2, and this is valid for a broad range of $< s_e >$ (*9*). Other explanation of the plot in Figure 2 has not been proposed for more than a decade since the first publication of this result (*10*). As its shape and

width are identical for various insulators we consider $\Theta(r)$ as an universal-type relation. It is a highly important possibility that $\Theta(r)$ can be measured by systematic experiments.

In a mathematical sense, there is another solution to the problem, as well. According to this, the $T(r,0)$ distributions may depend on individual MPs at any temperature except at $T = T_m - T_{ir}$ where $r = R_e$ in agreement with $\Theta(r)$, irrespective of MPs. However, as T_m is not a preferred temperature for the electronic energy deposition, it is evident that this mathematical solution has not any physical reality.

One may consider that when a model describes the track sizes correctly then it simultaneously describes the relationship in Figure 2 as well, as the correct track radii figure in both places. The problem is that the R_e–S_e track evolution is a simple smooth curve that can be given by several functions or a polynomial series, too. However, those are simply formal descriptions without any physical consequences for the ion-induced temperature. Therefore, the research should not be stopped for the investigation of the origin of this phenomenon that may lead to revealing a basically new effect.

There are several interesting ideas that have been proposed for explaining track formation in a particular solid or for some group of solids. Many different MPs are applied in these models. The reality of some of these models can be simply checked if the track radii follow the systematics shown in Figure 2. In the following sections, we analyze critically the reports on the predicted effect of some MPs on the formation of tracks.

2.2. Thermal diffusivity

This problem has been discussed already earlier (*11*). In the present paper, the key result is that the width of the $\Theta(r) = T(r,0)$ distributions does not vary with the composition in Figure 2. Therefore, the thermal diffusivities D must have close values in various insulators in that period of time when tracks are formed. Thus it is not a parameter leading to different track sizes. It is not reasonable in this respect referring to the values measured under equilibrium conditions. In Ref.(*11*) similar conclusion was drawn from the identical value of the Gaussian width $w = 4.5$ nm of the ion-induced temperatures in various insulators derived from the analysis of track evolution curves. Thus the analysis of the experimental results shows unambiguously that the thermal diffusivity is not a controlling parameter for track formation.

2.3. Velocity effect

It is a widely known experimental observation that low velocity ions induce larger tracks compared to high velocity ions (VE) (*2*). In spite of its relatively long history, the origin of this effect is not clarified yet. It is a reasonable assumption in the i-TS (Inelastic Thermal Spike) model that the higher the energy of the projectile, the broader the spatial distribution of the excited electrons leading to the reduction of the energy density and broadening of the distribution of the ion-induced temperature (*12*). In this model, the thermal energy is unchanged when the ion energy varies at $S_e =$ constant. The spatial distribution of electron energy is characterized by an MP, the absorption radius α_e which is the radius of a cylinder in which 0.66 of the electronic energy loss is stored on the target electrons (*12*). Then the variation of α_e with E – calculated by means of Monte Carlo simulations – is used for the

explanation of VE. High velocity ions induce broader electron energy distribution and lower energy densities leading to lower induced temperatures and smaller tracks.

The change of the absorption radius with E is suitable for following the variation of the track radius as required by VE. This is the element by which the VE is taken into account quantitatively in the i-TS model. It is an important condition that the projectile energy used in the calculation must be close to the experimental value otherwise the correction introduced by the absorption radius α_e will be incorrect.

Though the relationship between the variation of the absorption radius and VE seems to be a reasonable assumption, however, there is no evidence on the real broadening of the ion-induced temperature distribution with increasing E. On the other hand, it is evident that an additional MP apart T_m is in contradiction with the plot in Figure 2 and Equation (1) if it modifies the track size. As those are confirmed by experimental facts, this throws doubts on the validity of this model of VE. In a previous section, we have already applied the equation $\Theta(r) = T(r,0)$. Some important features are apparent in Figure 2. Actually, the experimental data show that the width of $T(r,0)$ is $w =$ constant, and it does not vary with the ion energy similarly to its shape. Thus the basic idea of the proposed explanation of VE is not confirmed by experiments (*12*). There are far-reaching consequences of this result concerning the possible mechanism of track formation. However, here we restrict our self only to the discussion of the VE.

In Figure 2, the shape, width and T_p are identical for $\Theta(r)$ at low and high ion velocities. The only difference is that at low ion velocities $< s_e > /3k = 3.42 \times 10^5\ \mathrm{nm^2\,K}$ ($E < 2$ MeV/nucleon) and at high ion velocities ($E > 8$ MeV/nucleon) $< s_e > /3k = 7.5 \times 10^5\ \mathrm{nm^2\,K}$. However, in spite of the different values of $< s_e >$ the peak temperature T_p and thermal energies ϵ are identical in both cases. It is remarkable, that the thermal energy ϵ is considerably lower than the deposited energy at low and high ion velocities, as well. Thus it is evident from the figure that it is not the whole deposited energy but only its $f < 1$ fraction is transferred to thermal energy. It follows from the above considerations that the efficiency f has different values at low and high ion velocities. The ratio $f_{LO}/f_{HI} = 3.42/7.5 \approx 0.46$ (see Figure 2). The values of the efficiencies can be simply estimated by applying the balance of energy to data in Figure 2 and using the $\rho c = 3Nk$ approximation where ρ and c are the density and specific heat. In a first step, the peak temperature is derived for a Gaussian distribution as

$$T_p = \frac{f\,\langle s_e \rangle}{3\pi k w^2} \tag{2}$$

Thus the complete expression for $\Theta(r)$ is

$$\Theta(r) = \frac{f < s_e >}{3\pi\, k w^2} \exp\{-r^2/w^2\}. \tag{3}$$

When the appropriate values of T_p and $< s_e > /3k$ are applied (see Figure 2), $f_{LO} \approx 0.38$, $f_{HI} \approx 0.17$ are obtained in good agreement with other results (*9*). Presently, the details of the transition between f_{LO} and f_{HI} are not known as there are no systematic experimental data available in the range 2 MeV/nucleon $< E <$ 8 MeV/nucleon.

Thus by analyzing the $\Theta(r)$ relationship it was found that only about 40% of the deposited energy is transferred to thermal energy at low ion velocities and this value is reduced further with the increase of E. The higher thermal energy ϵ means higher temperature and higher

energy density leading to the formation of larger tracks. This may be the origin of the VE. According to the results in Figure 2, the width of the temperature distribution does not change with E. Thus the variation of the absorption radius α_e is not related to VE, it does not lead simultaneously to broadening or narrowing of the temperature distribution.

In Ref.(13) the transfer of the energy of the electric field of the ionized lattice atoms to kinetic energy was estimated. It was found that it varies with E similarly to that observed for VE. It was concluded that the Coulomb explosion is a possible microscopic mechanism of VE. It was also found that the calculations were not sensitive to the actual form of the charge distribution. Thus this result is coherent with the plot in Figure 2.

2.4. Heat of fusion and track formation

When a system attains the temperature of phase transition a certain amount of energy, the so-called latent heat is absorbed or released by it. In the case of very fast thermal processes like ion irradiation, superheating may occur when the temperature may exceed the melting point without phase transition and it occurs at the superheating temperature. It is also a possibility, that the local electronic structure may be changed due to the intensive irradiation by an extent that the long-range order of the crystal is destroyed spontaneously without the deposition of the latent heat.

The heat of fusion L is an MP and usually it is equal to a considerable fraction of Q_m, the energy for heating up the solid to T_m from room temperature. The general proportionality between L and $T_m - T_{ir}$ is not realistic. Therefore, the plot in Figure 2 is useful in this case as well.

The plot in Figure 2 is a unambiguous evidence that track sizes depend only on a single MP – the melting temperature T_m. This is a clear instruction concerning the contribution of the heat of fusion L to track formation. It is important to point out that this is not the case of two physical assumptions with respect of the role of L and, *e.g.* the better agreement with the experimental data may suggest which of them is more realistic. Figure 2 and Equation (2) provide a simple mathematical condition for this problem. If R_e depended on L a smooth curve through the data in Figure 2 would be impossible and the theoretical calculations cannot lead to opposite results.

Actually, the following quantitative relation between R_{e1}, R_{e2} track radii and T_{m1}, T_{m2} melting temperatures can be simply derived for two solids, 1 and 2 from the enveloping curve $\Theta(r)$ in Figure 2 (9)

$$R_{e2}^2 - R_{e1}^2 = w^2 \ln\left(\frac{T_{m1} - T_{ir}}{T_{m2} - T_{ir}}\right) + w^2 \ln \frac{f_2 < s_{e2} >}{f_1 < s_{e1} >} \tag{4}$$

where the parameter $f = 0.17/0.4$ depends on the energy of the projectile. The validity of Equation (4) has been checked by reproducing accurately the ${R_e}^2$–S_e track evolution curves of insulators based on the track data of another insulator (9). This equation is very useful when the contribution of various MPs are studied. It shows most clearly the unique role of T_m for tracks compared to other MPs. In the following, we shall refer to this equation several times.

Thus the track radius R_{e2} can be obtained within experimental error having the necessary information on f and T_m. The most important point for the present problem is, that L

is not included in Equation (4), nevertheless, the track sizes can be estimated within experimental error using this equation. This is supported by our previous results when the R_e^2–S_e track evolution curve was calculated for $NiFe_2O_4$ based on the data for $Y_3Fe_5O_{12}$ in good agreement with the experimental data. Similar successful calculation was performed for the TeO_2 – mica pair with ignoring the heat of fusion again (*9*). Such calculations can be repeated for any insulator in Figure 2 including $LiNbO_3$ as well as it can be found in both lists in the figure.

The above considerations seem to be rather convincing evidence that *L* may have only a minor contribution to track formation in $LiNbO_3$. On the other hand, there is a study where including the tabulated value of *L* into the calculations leads to significant improvement of the agreement with the experiments on track evolution in $LiNbO_3$ (*14*). Thus the heat of fusion seems to have a considerable contribution to track formation and this view is widely accepted, though no similar analysis has been reported for any other solid. These two results for $LiNbO_3$ are obviously in contradiction. A peculiarity of this problem is that the same database is used in both analyses leading to opposite conclusions while only one of the two considerations may be valid. This is not a particular question as the effect of various MPs on track formation is a fundamental problem. This problem is related to the proceeding of a phase transition under equilibrium or sharply non-equilibrium conditions. Therefore, the solution may have interesting theoretical consequences, as well.

In Ref.(*14*) track evolution in $LiNbO_3$ was analyzed. Compared to an earlier analysis in Ref.(*15*) three tracks induced by U and Gd ions showing considerable deviations were ignored (compare Fig. 5b and 12 in Ref.(*14*) to Fig. 2 in Ref.(*15*)). On the other hand, three tracks were included without any information on the irradiation conditions. However, they need not to be ignored for this reason as they can be identified using Refs. (*16*, *17*) as tracks induced by high energy Ni ion irradiations.

Finally, tracks produced by ions (in brackets) with $E = 9.7$, 5.9, 2.8, (Ni) 9.0, 2.7, (Sn) 2.6, (Gd) 2.01 and 2.73 (U) MeV/nucleon were analyzed in Ref.(*14*). A smooth R_e–S_e curve was obtained by this procedure. We note that in the case of accurate measurements it would not be possible drawing a smooth curve through track data with a similar broad range of the variation of *E*. The separation of the curves of $Y_3Fe_5O_{12}$ in Figure 1 well demonstrates the scale of the effect of the ion velocity. As there is an overall agreement concerning VE, a similar separation is expected in $LiNbO_3$ for tracks induced by 2.8 (Ni) and 2.7 (Sn) MeV/nucleon and tracks induced by 9.7 (Ni) and 9.0 (Sn) MeV/nucleon, respectively. The unexplained smooth R_e–S_e curve and ignoring some data with high deviations may lead to the reduction of the confidence in the results of this analysis.

In Ref.(*14*) the analysis was made applying the i-TS model (*12*). Usually, this model provided good agreement with the experiments. In the given case it was accepted by the authors as if $E \approx 10$ MeV/nucleon were valid for all tracks (see Fig. 12b in Ref.(*14*)) though $E \approx 2.5$ MeV/nucleon for the majority irradiations. Compared to the real E values this was apparently an arbitrary step and no explanation was given by the authors. In Figure 3, the original figure in Ref.(*14*) is completed with the appropriate values of *E*.

Initially, the results showed large separation of the experimental and the theoretical curve calculated with $L = 0$ (see the dashed line in Figure 3). However, this difference disappeared after taking into account the supposed contribution of *L*. The authors considered this as a strong evidence confirming the contribution of *L* to track formation (*14*).

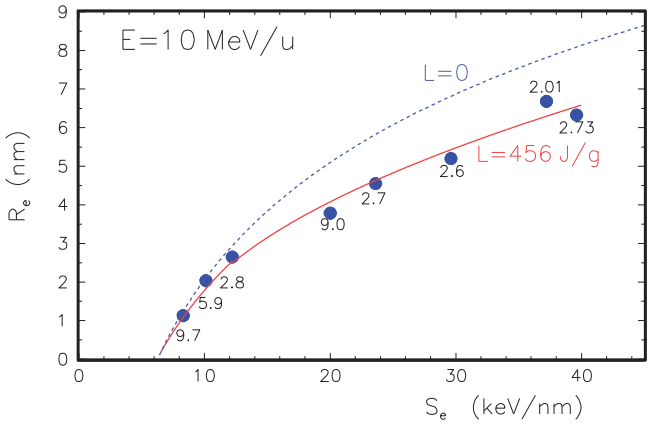

Figure 3. Calculation of the R_e track radii for $LiNbO_3$ applying the i-TS model (*14*); *L* – heat of fusion. The numbers at the track radii are the specific ion energies in MeV/nucleon units (*14, 15*).

However, there are several problems with this consideration. We refer here to the warning of Toulemonde et al. in Ref.(*12*) 'To avoid misinterpretation, any data comparison should (therefore) only be performed within the same velocity range.' This is a clear physical condition that is appropriate for this case, as well. Obviously, this was not followed in Ref.(*14*) by the authors.

Usually, a mathematical agreement between experiments and a theoretical calculation is considered to be sufficient for justifying the validity of the assumptions. A peculiarity of this situation is that because of the rather different *E* values in the experiment and in the calculation, the good agreement between the experimental and the calculated curves does not justify the contribution of *L* because of the VE. The situation is evident when looking at the curves in Figure 1 for $Y_3Fe_5O_{12}$. The coincidence of the LO and HI curves in Figure 2 well demonstrates that VE has a similar magnitude in a number of insulators including $LiNbO_3$ and $Y_3Fe_5O_{12}$. The theoretical calculations for $E = 10$ MeV/nucleon would follow approximately the HI curve in Figure 1 for $Y_3Fe_5O_{12}$ while track evolution with $E = 2$–3 MeV/nucleon must be close to the LO curve (see Fig. 11 in Ref.(*12*) for more details). The difference between track radii induced by projectiles with $E \approx 10$ and 2.5 MeV/nucleon may exceed 40% for the same values of S_e. This means that while there seems to be a complete mathematical agreement for $E = 10$ MeV/nucleon when an R_e–S_e curve is calculated with the contribution of *L* in $LiNbO_3$, the estimate for the realistic $E = 2.5$ MeV/nucleon may exceed considerably even the curve with $L = 0$ in Figure 3. According to the words of Toulemonde, it is evident that the calculation for $E \approx 2.5$ MeV/nucleon is closer to the correct result for the data in Figure 3. Thus the excellent agreement demonstrated in Ref.(*14*) for $E = 10$ MeV/nucleon is misleading, it is rather a counter argument for the contribution of *L*.

We note that if the above method were accepted then it would be simple to prove in another case that only a fraction of *L* ought to be taken into account for ion-induced melting. The only change in the proof would be applying $E < 10$ MeV/nucleon in the calculations; *e.g.* $E \approx 6$ MeV/nucleon for a 50% reduction of *L*.

There is another serious weak point of the paper even if an agreement would have been found for $E = 2.5$ MeV/nucleon. The calculation is based on the improvement of a particular

fitting to some data points applying one of the proposed models. This is not a guaranty that the contribution of L is necessary indeed for a correct description of the effect. There are other proposed functions for fitting (Analytical Thermal Spike Model (*1*), square root function (*12*)) where the supposed contribution of L does not lead to an improvement. Thus the reliability of the evidences in support of the role of L is rather doubtful in Ref.(*14*). However, the most powerful counter argument is Equation (4) as track radii can be estimated within experimental error without any information on the heat of fusion. This is fulfilled well for all insulators listed in Figure 2.

In summary, the track data used in Ref.(*14*) are not suitable for a common thermal spike analysis because of the selection of the track data and a broad range of variation of E in $LiNbO_3$. The good formal agreement between the calculated and experimental track values for $E = 10$ MeV/nucleon is just a proof to the contrary of the author's claim because of the contradiction to the consequences of VE. Additionally, even a formally correct verification were not sufficient as it would be valid only for one of the models and do not for the phenomenon. Thus after thorough reconsidering the analysis in Ref.(*14*) it was concluded that those results were not suitable throwing any doubt on the minor contribution of L to track formation.

The contribution of L to track formation is not a simple theoretical problem. It has significant practical consequences, as well. In a model, where L is ignored, track formation starts immediately when the local temperature is $T = T_m$ at $S_e = S_{et}$. In a model which takes into account this contribution, track formation starts after the deposition of an additional energy L at $T = T_m$. As S_{et} is determined by the experiments its value cannot differ much in the two approaches and we take them equal for simplicity. Thus when L is not ignored, $T = T_m$ is attained at about $S_e \approx S_{et}/(1 + L/Q_m)$, where Q_m, is the energy for heating up the solid to T_m from room temperature. This S_e value is considerably lower than S_{et} while $T = T_m$ at $S_e = S_{et}$ for $L = 0$. Now, the calculation of the induced temperatures is not correct in one of the two approaches. Our previous considerations show that $L = 0$ during the process of track formation. Therefore, the calculation of the induced temperatures are not reliable, they are considerably overestimated in all those cases when the contribution of L is taken into account. As a result, the error in estimating the induced temperature may exceed 50% for Gd_2TiO_7 or UO_2. This may have important practical consequences as the stability of irradiated solids are intensively affected by thermal processes.

In this paper, we analyzed opposing results regarding the effect of L on the induced track radii. It is shown that track radii R_e do not depend on the heat of fusion L. Obviously, the validity of these results is limited by the experimental error of the measurements of track radii, that is usually 5–10%.

2.5. Band gap energy

It was assumed in Ref.(*12*) that the band gap energy E_g may have a rather indirect effect on the track size as it may affect the value of the electron mean free path λ that is the control parameter in the i-TS model. Thus E_g may change indirectly the track size thus it may be another MP besides L controlling the track radius. We discuss this problem in detail because similarly to the case of $LiNbO_3$ the same experimental data have been used for supporting the minor role of E_g and also for its significant contribution to track formation. The clarification of this problem deserves a thorough analysis.

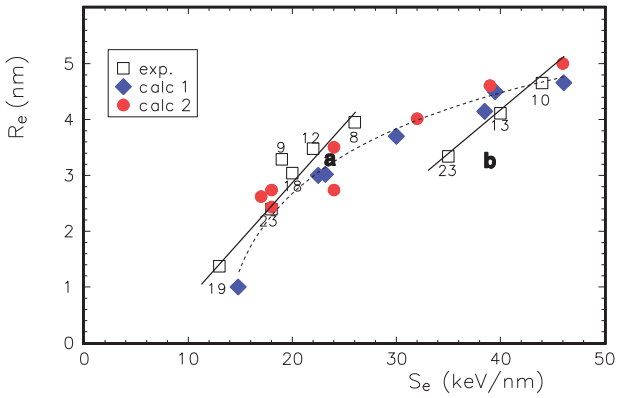

Figure 4. Variation of the R_e track radii in $BaFe_{12}O_{19}$ (*6*, *7*). The solid lines (a) and (b) were calculated applying the i-TS model (*18*). The numbers at the track radii are the specific ion energies in MeV/nucleon units. R_e values for $BaFe_{12}O_{19}$ were calculated based on tracks in TeO_2 (*5*) (calc1) and in $NiFe_2O_4$ (*1*) (calc2) applying Equation (4); the dashed curve is a fit to the calculated track values.

A prominent example of the effect of E_g is the track formation in $BaFe_{12}O_{19}$ where $E_g = 1$ eV which is much lower than in other insulators in Figure 2. A theoretical basis is given by the i-TS model. A detailed analysis of experimental data has been given only for $BaFe_{12}O_{19}$ in this respect (*18*), nevertheless, the result on the effect of E_g on track formation is widely accepted for other solids, as well. Tracks induced in $BaFe_{12}O_{19}$ by HI ions are shown in Figure 1. We emphasize that tracks along the curve (a) show quite similar behavior to other insulators listed in Figure 2. We remind that all those track radii can be calculated within experimental error by using only T_m as an MP**.** Moreover, E_g varies in Figure 2 in the range 1–12 eV. The smooth enveloping curve in Figure 2 excludes the possibility for those insulators including $BaFe_{12}O_{19}$ that E_g might have any effect on the track size. We emphasize that the data of $BaFe_{12}O_{19}$ are also included in this figure demonstrating the existence of a relationship between tracks in various insulators that would not be possible with the contribution of MPs apart T_m. At the same time, the same data in $BaFe_{12}O_{19}$ represent the main argument in Ref.(*18*) justifying the contribution of E_g to track formation. Below we reconsider the experimental data and the analyses.

The experimental data for $BaFe_{12}O_{19}$ are shown in Figure 1. The samples with a thickness of about 50–60 μm were irradiated in stacks by ions Mo, Xe, Pb and U ions in the GeV range (*6*, *7*, *19*, *20*). Track radii were measured by applying Mössbauer spectrometry. Unlike all other R_e–S_e curves, there is a significant drop in track radii that appears at $S_e > 34$ keV/nm in this solid though the R_e^2–S_e track evolution curves are smooth and continuous in other track forming insulators without exceptions.

In Ref.(*18*) the analysis was based on the assumption that VE led to the separation of the track evolution curves in $BaFe_{12}O_{19}$. Theoretical lines were calculated for two beam energies $E = 7$ and 23 MeV/nucleon applying the i-TS model with fitting parameters $\lambda = 6.9$ and 9.5 nm and the results are shown in Figure 4. An excellent agreement with the experimental data along lines (a) and (b) in Figure 1 was obtained with these parameters.

The values of λ were considerably higher than those in other insulators ($\lambda = 4$–5 (*12*)) and this was attributed qualitatively to the effect of the low band gap energy. This is the main evidence for the assumed λ–E_g relation.

For resolving the discrepancies between the conclusions of the two approaches, the analysis in Ref.(*18*) was completed with the missing mean values of the ion energies E along the thickness of the samples, by using the original publications (*6*, *7*, *19*, *20*). Estimates were obtained using the values of the thickness of the samples, the kind of the projectiles and their initial energies and also the average S_e values given in Refs.(*6*, *7*, *19*, *20*). The results are shown in Figure 4. As the experiments were published by 1991 we applied the TRIM91 code in our calculations.

In Figure 4 in Ref.(*18*), the induced temperature was calculated for Xe irradiation with a mean energy $E = 7$ MeV/nucleon and $S_e = 26$ keV/nm. In our calculations a slightly higher value $E = 8$ MeV/nucleon was estimated under the same conditions. In spite of some uncertainties in the parameters of the irradiated samples, the small difference characterizes the reliability of our estimates. After considering all estimated values of E it became evident that $E = 7$ and 23 MeV/nucleon applied in the theoretical calculation are too far from the real conditions. Thus similarly to the case of $LiNbO_3$, the two values of E used in the analysis in Ref.(*18*) were not coherent with those of the mean ion energies applied in the experiments.

Consequently, the analysis in Ref.(*18*) was made at arbitrary values of E that were very different from the real ones. This is in contradiction with the conditions required for a reliable analysis, since it distorts the calculated track sizes as it has been shown in the previous section for $LiNbO_3$. The demonstration of the good fit with two arbitrary values of E is only seeming, not a real argument. Actually, this cannot serve as an evidence supporting the validity of any statement for $BaFe_{12}O_{19}$ including the effect of E_g on track radii. This may have further consequences, however, it is beyond the scope of this paper. Opposite to the conclusion in Ref.(*18*), the separation of the track radii cannot be related to the projectile velocities as the average values of E along the curves (a) and (b) are very close: $E = 14.8$ and 15.3 MeV/nucleon. All these problems were hidden as the values of E were not given for the projectiles in Ref.(*18*). The situation is similar to that with $LiNbO_3$ in the previous section. Inappropriate values of the specific ion energies devaluate the results of the theoretical analysis.

However, any of the above considerations do not explain the drop of the track evolution curve for about $S_e > 34$ keV/nm. In Section 2.1, we mentioned that Figure 2 demonstrates that there must be a simple quantitative relationship between track radii measured in different insulators. Equation (4) is an equation that can be applied for the calculation of a track evolution curve of an insulator using the data of an other solid. It is remarkable that apart from T_m no other MPs are involved in Equation (4). In Ref.(*9*) we have already demonstrated that this method works very accurately for the TeO_2–mica and the $Y_3Fe_5O_{12}$-$NiFe_2O_4$ pairs. Here we use the same procedure for $BaFe_{12}O_{19}$ as it also exhibits common features with other insulators in Figure 2. We use the LO and HI track data of TeO_2 (*5*) and $NiFe_2O_4$ (*1*), respectively for estimation. The details are given in Ref.(*9*) and the results are shown in Figure 4.

The calculated curve clarifies considerably the situation. Track radii derived from the data of TeO_2 and $NiFe_2O_4$ are in good agreement. The overwhelming majority of the estimated and real track radii are in reasonable agreement for $BaFe_{12}O_{12}$. Some of the larger than the usual deviation may be the consequence of the large thickness of the samples ($\sim$50–60 μm) leading to much higher than usually uncertainties of R_e, and S_e. Additionally, the mean E values do not characterize unambiguously the effect of the ion velocity. Please remember that the difference between track radii may be about 40–50% in the

LO and HI regimes for S_e = constant. For example, the upward deviation of tracks size at $S_e = 19$ keV/nm $S_e = 26$ keV/nm ($E = 9$ and 8 MeV/nucleon, respectively) is conceivable as in about half of the sample thickness (back half) ($\sim 30\,\mu$m) approximately $8 > E > 2$ MeV/nucleon is valid. Thus as a consequence of the VE much larger tracks are formed in the back half than in the front half of the samples with $E > 8$ MeV/nucleon while track evolution is nominally in the HI range in the whole sample. This may lead to the formation of larger tracks in average than it would be expected from an irradiation in the HI range.

We have shown by making use of the scaling features of the R_e^2–S_e track evolution that these curves are in good agreement for various insulators listed in Figure 2 (*3*). As that common curve is represented by the dashed line in Figure 4, therefore, the plot in Figure 4 also demonstrates that the same is valid for $BaFe_{12}O_{19}$ as well. According to the above analysis the low value of E_g in $BaFe_{12}O_{19}$ does not lead to marked changes in the track evolution. It is not possible to notice any consequence of the low value of E_g.

Thus the comparison of the calculated curve with the experimental data shows that the track evolution curve of $BaFe_{12}O_{19}$ and other insulators in Figure 2 are coherent. The only problematic track is at $S_e = 35$ keV/nm. All other data are in reasonable agreement with the estimated curve which demonstrates the common behavior of these insulators. Our opinion is, that a single track radius that was not measured under well-defined conditions cannot provide the necessary experimental basis for proving an exceptional behavior of a solid. Thus it is reasonable accepting that presently there are no reliable data supporting the existence of such drop in $BaFe_{12}O_{19}$.

The problem of the possible drop of track radii in $BaFe_{12}O_{19}$ is a good example demonstrating the advantageous consequences of the scaling features of tracks in a number of insulators. Otherwise only a complete repeated experiment could answer this question. This emphasizes the importance of performing analyses making use of quantitative results on other insulators. On the other hand, while there are practical advantages of using thick samples, the uncertainties in the experimental data on E, R_e and S_e due to VE may reduce the value of these experiments considerably.

We note that the λ–E_g relationship was also supported by a second study where track evolution in a high-T_c superconductor (HTCS) was analyzed using the i-TS model (*21*). There is a principal problem: the i-TS model is based on the quasi-free electron model (*22*) which is not suitable for describing the processes in highly anisotropic solids like HTCSs or GeS. Thus the reliability of the result of that analysis is rather dubious.

Thus the analysis of the original experimental data does not provide reliable evidences that the band gap energy E_g affects the track size. Thus these results do not throw doubts on the validity of the conclusion that individual MPs apart from T_m are inefficient for modifying the track sizes in insulators.

2.6. Final remarks

It is rather unexpected that such a universal-type relation like Equation (4) may exist between the results of independent experiments. As these insulators all have been selected for experiments 'randomly' regarding the final results in Figure 2, we assume that a considerable number of insulators may exhibit similar features exhibiting independence on MPs. Even up to now, tracks were induced in about 3–4 times more insulators compared to those included in Figure 2. However, our analysis requires the accurate knowledge of S_e, T_m, E, and

of the value of the efficiency *f*. Because of this last condition, we cannot use the results of experiments performed in the range $8 > E > 2$ MeV/nucleon where f varies in the range $0.4 > f > 0.17$.

The significance of the identical behavior is not limited to the simplification of the description of the processes. The consequences extend to the temporal and spatial variation of the temperature and deformation fields and to their effects on the local microstructure. Besides ion-induced track formation the theories of inelastic sputtering, mixing, swelling, formation of surface hillocks are also concerned. The results draw attention to the possibility that the equilibrium parameter values may change considerably under spike conditions.

Equation (3) describes a behavior in insulators which does not depend on the kind of components and the compositions. This is a unique phenomenon in physics. Moreover, several very intensive and complicated physical effects are excited simultaneously during bombardment by swift heavy ions. This phenomenon has not been described or predicted by any model and simulation studies. It is assumed that the underlying mechanism may be substantially different from those which have been applied previously in the theories of radiation effects. Thus the investigation of this effect may lead to a basically new knowledge.

3. Conclusions

A common quantitative relationship exists between track radii R_e and the melting temperatures T_m whose validity is demonstrated for 14 insulators including $LiNbO_3$ and $BaFe_{12}O_{19}$. This is attributed to the consequence of the common induced temperature $\Theta(r) = T_p \exp\{-r^2/w^2\}$ so that $r = R_e$ for $\Theta(r) = T_m - T_{ir}$ for $< s_e > /3k =$ constant in various insulators. By using the $\Theta(r)$ relation it is shown for insulators that thermal diffusivity D, the heat of fusion *L*, the band gap energy E_g and the absorption radius α_e have no effect on the track size as track radii depend only on a single MP – T_m. This is an evidence supporting the validity of the thermal spike mechanism. By applying $\Theta(r)$ track radii have been reproduced quantitatively for a number of insulators using only T_m as an MP.

Earlier experiments on $LiNbO_3$ and $BaFe_{12}O_{19}$ with opposite results concerning *L* and E_g are critically discussed. A specific feature of the problem is that the same track data are used for supporting opposite conclusions. The track data in $LiNbO_3$ and $BaFe_{12}O_{19}$ were completed with the appropriate values of *E* missing from the original analyses. A common deficiency of those studies was that track data were rather mixed regarding the ion velocities and arbitrary values of *E* were used in the analyses. However, agreement with the experiments is meaningful only when close values of *E* are applied in the calculations and the experiments. As this is not fulfilled, the excellent mathematical agreement demonstrated by the authors for $LiNbO_3$ and $BaFe_{12}O_{19}$ are considered as the result of occasional coincidences, they cannot serve as an evidence. As no more studies on other materials have been reported, therefore, the conclusions on the contribution of *L* and E_g to track formation remain experimentally unfounded. The experimental data in Figure 2 show unambiguously that the parameters of the ion-induced temperature distribution does not vary with the ion velocity. Therefore, the origin of the VE is not related to the absorption radius α_e as it is widely accepted.

Disclosure statement

No potential conflict of interest was reported by the author.

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